ASSIGNMENT-3

enumerate() Function

Definition

enumerate() is a built-in Python function that adds a counter (index) to an iterable (like list, tuple, string) and returns it as an enumerate object.

- Syntax

enumerate(iterable, start=0)

- iterable → list, tuple, string, etc.
- start → starting index (default = 0)

Example 1: Basic Usage

```
names = ["Geethika", "Ravi", "Anil"]

for index, name in enumerate(names):
    print(index, name)
```

Output:

```
0 Geethika
1 Ravi
2 Anil
```

Example 2: With Custom Start Value

```
for index, name in enumerate(names, start=1):
    print(index, name)
```

Output:

```
1 Geethika
2 Ravi
3 Anil
```

---

isinstance() Function

Definition

isinstance() checks whether an object belongs to a specific class (type).

🔹 Syntax

```
isinstance(object, classinfo)
```

- Returns True or False

Example 1: Basic Type Check

```
x = 10

print(isinstance(x, int))     # True
print(isinstance(x, str))     # False
```

Example 2: Multiple Types

```
x = 10.5

print(isinstance(x, (int, float)))   # True
```

Example 3: Custom Class

```
class Student:
    pass

s = Student()

print(isinstance(s, Student))   # True
```

Garbage Collection (GC) in Python

Definition

Garbage Collection (GC) automatically frees memory by deleting unused objects.

Python mainly uses:

1. Reference Counting
2. Generational Garbage Collection

To work with GC manually, we use the gc module.

Import GC Module

import gc

Important GC Functions

---

gc.collect()

Forces garbage collection manually.

import gc

gc.collect()

Returns number of objects collected.

---

gc.enable()

Enables automatic garbage collection.

gc.enable()

---

gc.disable()

Disables automatic garbage collection.

gc.disable()

---

gc.get_count()

Returns current collection counts.

```
print(gc.get_count())
```

---

gc.get_threshold()

Returns GC thresholds.

```
print(gc.get_threshold())
```

---

gc.set_threshold()

Sets new threshold values.

```
gc.set_threshold(700, 10, 10)
```

Example of Garbage Collection

```
import gc
class Test:
    pass

t1 = Test()
t2 = Test()
del t1
del t2
collected = gc.collect()
print("Objects collected:", collected)
```